

DeepMutate-3D

DeepMutate-3D: interactive protein language model mutation scanning on predicted structures

Rashid Karim

Independent Researcher, Cambridge, MA, USA

Correspondence: rashidsaadman@gmail.com

Code: <https://github.com/Saadman/DeepMutate-3D> Web application:
<https://huggingface.co/spaces/ras1992/DeepMutate-3D>

Abstract

Motivation. Protein language models predict the functional effect of amino acid substitutions without training on experimental data, but their output is a table of numbers. The structural context that makes such a table interpretable, which residues sit in an active site, a buried core, or a flexible loop, lives in separate software. Bridging the two currently requires local installation, model weights, and scripting, which places the method out of reach of many laboratory scientists.

Results. DeepMutate-3D scores all 19 possible substitutions at every position of a protein using ESM-2 log-likelihood ratios, condenses them into a per-residue sensitivity score, retrieves the matching AlphaFold model, and paints the scores onto the structure as an interactive heat map in a web browser. Predictions are zero-shot. On the ProteinGym substitution benchmark (217 assays, 696,311 measured variants) the deployed configuration reaches a mean Spearman correlation of 0.425, consistent with the published ESM-2 650M baseline of 0.414. On 38,901 ClinVar variants across 2,011 proteins it separates pathogenic from benign substitutions with a mean per-protein AUROC of 0.881. In three case studies with independent ground truth it recovers TP53 cancer hotspots (permutation $p = 0.0007$), enzyme active sites, and all eight disulfide-bonded cysteines of lysozyme C (hypergeometric $p = 1.0e-11$).

Availability. Apache-2.0. Runs in a browser with no installation, on shared NVIDIA hardware, or locally on CPU or Apple Silicon.

1. Introduction

Determining which residues in a protein tolerate mutation is a recurring question in variant interpretation, enzyme engineering, and functional annotation. Deep mutational scanning answers it experimentally but requires months of work per protein. Computational predictors offer a fast alternative, and protein language models in particular have proven capable of zero-shot variant effect prediction: trained only to reconstruct masked residues across evolutionary sequence data, they implicitly learn the constraints that shape protein families [1, 2].

The practical bottleneck is not accuracy but access and interpretation. A predictor returns a score per substitution. Interpreting that score requires knowing where the residue sits in the fold. A position scoring as highly constrained is only actionable once one knows it lines a catalytic pocket rather than a disordered terminus. Structural prediction has become universally available [3, 4], yet in practice the score and the structure are produced by different tools and joined by hand.

DeepMutate-3D closes that gap in a single interface. It is not a new scoring method. The underlying computation is the masked-marginal log-likelihood ratio of Meier et al. [2] applied to ESM-2 [1]. The contribution is integration and accessibility: a scan that requires no installation, no model weights, and no scripting, returning a structure that can be rotated and interrogated residue by residue, together with exportable tables and coordinate files.

Several established predictors address variant effect prediction, including alignment-based and language-model approaches evaluated in ProteinGym [5], and web servers exist for structure-based stability prediction. DeepMutate-3D does not compete with these on accuracy, and no head-to-head comparison against them is attempted here. Its distinction is the combination of zero-shot scoring, automatic structural mapping, and browser-based access with no installation.

Because the tool wraps an established method, its value depends on the wrapper being faithful. This note therefore reports validation on two independent benchmarks, an ablation across model sizes, a comparison of scoring protocols, and three case studies against ground truth that was established independently of this work.

2. Implementation

2.1 Scoring

For a sequence of length L , position i , and candidate residue m , the model computes a log-likelihood ratio against the wild-type residue:

$$\text{LLR}(i, m) = \log P(m \mid \text{context}) - \log P(\text{wt}_i \mid \text{context})$$

Values near zero indicate that the model finds the substitution as plausible as the wild-type residue; strongly negative values indicate a substitution the model regards as unlikely given the evolutionary context. Because the quantity is a ratio, it is insensitive to positions at which the model is uniformly uncertain.

The 19 non-wild-type ratios at each position are averaged into a single Residue Sensitivity Score:

$$S_i = (1/19) * \text{sum over } m \neq \text{wt}_i \text{ of } \text{LLR}(i, m)$$

Two scoring protocols are provided. *Masked marginals* replaces position i with a mask token before each forward pass, following Meier et al. [2], and costs one forward pass per residue. *Wild-type marginals* uses a single forward pass over the intact sequence, reading the output distribution at every position. Three ESM-2 sizes are selectable (35M, 150M, 650M parameters).

2.2 Sequences longer than the model context

ESM-2 has a 1,024-token context, which after special tokens accommodates 1,022 residues. Longer proteins are covered with overlapping windows of 1,022 residues at a stride of 766. Log-probabilities are accumulated in sequence space and divided by a per-position visit count, so that every interior residue is scored in two different contexts and averaged. The overlap avoids the systematic error that arises when a residue falls at a hard window boundary with no downstream context.

2.3 Structure retrieval and mapping

The predicted structure is retrieved from the AlphaFold Protein Structure Database [4] by UniProt accession, queried through the prediction API so that the current model version is used rather than a hard-coded filename. Sensitivity scores are written into the B-factor column of the PDB record, replacing pLDDT, which allows 3Dmol.js to colour the cartoon representation by a numeric property without server-side rendering. The colour scale is a red-white-blue gradient with symmetric limits, placing the neutral midpoint at exactly zero so that colour is comparable across proteins. Exported coordinate files carry REMARK records documenting that the B-factor column no longer contains pLDDT.

2.4 Interface and deployment

The application is a single-file Gradio program. Hovering any point on the structure reports the residue's sensitivity score, its categorical verdict, and its most damaging and most tolerated substitutions. Results are exportable as a CSV table and as a scored PDB file that opens directly in PyMOL or ChimeraX.

The deployed instance runs on Hugging Face Spaces using dynamically allocated NVIDIA hardware, with GPU time reserved per request in proportion to the size of the job. The same source runs unmodified on CUDA, Apple Silicon, or CPU.

3. Validation

All evaluations are zero-shot. No experimental or clinical measurement is shown to the model at any point.

3.1 Mutation effect prediction

ProteinGym [5] aggregates deep mutational scanning experiments into a common benchmark. Predictions were generated for every measured single substitution and correlated with experimental fitness per assay using Spearman's rho.

Table 1. ProteinGym substitution benchmark, 217 assays, 696,311 variants, wild-type marginal scoring.

Model	Mean rho	Median rho	rho > 0.3	rho > 0.5	Compute
ESM-2 35M	0.291	0.321	51%	19%	17 s
ESM-2 150M	0.383	0.430	70%	35%	19 s
ESM-2 650M	0.425	0.477	77%	44%	47 s

The deployed configuration (650M) reaches 0.425. Published values for ESM-2 650M on this benchmark are in the region of 0.41 to 0.42. The present figure is therefore in the expected range, but it should not be read as outperforming those reports: scoring protocol, assay filtering, and aggregation scheme all differ between published evaluations, and the comparison here is indicative rather than exact. The purpose of Table 1 is to establish that the wrapper does not degrade the underlying method, not to claim an improvement over it.

3.2 Scoring protocol

Masked and wild-type marginal scoring were compared pairwise on an identical subset of 197 assays at 150M parameters.

Table 2. Paired comparison of scoring protocols, 197 assays, ESM-2 150M.

Protocol	Mean rho	Median rho	Compute
Wild-type marginals	0.392	0.441	11 s
Masked marginals	0.399	0.453	4,903 s

Masked marginals is superior on 72.1% of assays, but the paired mean difference is +0.0069 (95% CI +0.0050 to +0.0089) for 434 times the compute. The theoretically preferable protocol therefore buys a real but very small improvement at large cost.

Read alongside Table 1, this yields the practically useful observation that compute is better spent on model capacity than on scoring protocol: ESM-2 650M under wild-type marginals (rho 0.425) exceeds ESM-2 150M under masked marginals (rho 0.399) while using two orders of magnitude less computation. This comparison was performed at 150M; the corresponding measurement at 650M was not made, and the claim is not generalised beyond the model size tested.

3.3 Clinical variant classification

An independent task on independent data. Using the ClinVar substitution set curated within ProteinGym, variants of each protein were ranked by predicted damage and scored by AUROC against their pathogenic or benign annotation.

Table 3. ClinVar classification, 2,011 proteins, 38,901 variants (22,669 pathogenic), ESM-2 650M.

Metric	Value
Mean per-protein AUROC	0.881
Median per-protein AUROC	0.931
Proteins with AUROC > 0.7	89.4%
Pooled AUROC, per-protein z-scored	0.771
Pooled AUROC, raw scores	0.868

Per-protein AUROC is the primary metric because it matches the clinical question: given the variants observed in one gene, which are most likely to be damaging. Raw scores are not comparable between proteins, so pooled analysis requires per-protein standardisation; the pooled figure is lower than the mean per-protein figure because pooling mixes proteins of differing discriminability. Section 5 discusses the limitations of ClinVar as a benchmark.

3.4 Case studies

TP53 cancer hotspots. The six most frequently mutated positions in TP53 across human tumours have a median sensitivity of -7.89 against a protein-wide median of -2.13, and a median constraint percentile of 92.5. Against 20,000 random six-position draws from the whole protein, $p = 0.0007$.

That null is however too permissive, because TP53 carries long disordered terminal regions that are trivially unconstrained, and all six hotspots lie within the DNA-binding domain. Restricting the null to that domain (residues 94 to 292, median sensitivity -3.74) gives $p = 0.009$. The effect survives the stricter test, but is an order of magnitude weaker than the whole-protein comparison implies, and the restricted value is the one that should be cited.

Enzyme active sites. Six enzymes were selected before scoring on two criteria: the presence of curated UniProt active site or binding site annotations, and a length within the model context so that no windowing was required. No enzyme was examined and then discarded. Across the panel, annotated residues reach a median constraint percentile of 86.8. Four of six enzymes reach $p < 0.01$ by permutation (TEM-1 beta-lactamase $p = 0.0001$, phosphoglycerate kinase 1 $p = 0.00005$, GAPDH $p = 0.00015$, carbonic anhydrase 2 $p = 0.005$). Two do not: lysozyme C (three annotated sites, insufficient for significance) and cationic trypsin ($p = 0.081$), whose annotation set mixes substrate-binding with catalytic residues. Binding residues are under weaker evolutionary constraint than catalytic ones, and this distinction is visible in the result.

Lysozyme disulfide bridges. Of the ten most constrained positions in lysozyme C, eight are cysteines, and they are exactly the eight residues UniProt annotates as forming the protein's four disulfide bridges (C24-C145, C48-C133, C82-C98, C94-C112). Drawing ten positions at random from 147, the probability of capturing all eight is $1.0e-11$. The remaining two positions are a buried glycine and a conserved tryptophan in the substrate-binding cleft. The model received sequence alone, with no structure, no annotation, and no experimental data.

This result requires one qualification. Cysteines are constrained as a class: across the panel, cysteine positions reach a median constraint percentile of 96.6 in lysozyme and 90.2 in trypsin against a non-

cysteine median near 48, so part of the effect is generic rather than specific to disulfide bonding. A partial control is available in carbonic anhydrase 2, whose single cysteine does not participate in a disulfide bond and reaches only the 40.5 percentile, below the protein median. This suggests the model distinguishes bonded from free cysteines rather than merely conserving the residue type, but with a single control this remains suggestive rather than established.

3.5 Runtime

Table 4. Inference time, ESM-2 650M, identical proteins. CPU and Apple Silicon measured on an Apple M3 (6 threads for CPU); NVIDIA figures from dynamically allocated Hugging Face hardware, taking the minimum of repeated calls to exclude allocation overhead.

Protein	Length	Protocol	M3 CPU	M3 GPU (mps)	NVIDIA
Haemoglobin alpha	142	wild-type	0.41 s	0.09 s	0.18 s
Haemoglobin alpha	142	masked	30.98 s	7.82 s	1.45 s
p53	393	masked	236.19 s	62.27 s	10.99 s
SARS-CoV-2 spike	1,273	wild-type	3.50 s	0.43 s	0.31 s
SARS-CoV-2 spike	1,273	masked	not run	not run	152.33 s

Wild-type marginal scoring is fast on all hardware; the complete 1,273-residue spike protein is scored in under four seconds even on CPU. GPU acceleration matters for masked marginal scoring, where it delivers a 5.4 to 5.7-fold speed-up at matched protein length and where the advantage grows with length because attention cost scales quadratically. On small jobs under the fast protocol, dynamically allocated GPUs are slower than a laptop, because a fixed allocation overhead of approximately one second dominates.

4. Use

A scan requires a UniProt accession. Leaving the sequence field empty causes the canonical sequence to be retrieved automatically, which guarantees that scores and structure share residue numbering. Outputs are an interactive structure, a per-residue table with the most damaging and most tolerated substitution at each position, and CSV and PDB downloads.

Typical applications are prioritising positions for mutagenesis, identifying candidate functional sites in uncharacterised proteins, selecting tolerant positions for library design, and providing structural context for variants of uncertain significance.

5. Limitations

The method is not novel. Scoring is the established masked-marginal procedure of Meier et al. [2]. Section 3.1 reproduces rather than improves on published performance.

ClinVar is an imperfect benchmark. Pathogenic annotations are enriched in conserved functional domains and benign annotations in tolerant regions, which inflates apparent performance relative to prospective clinical use. Some ClinVar submissions also incorporate computational predictions, introducing partial circularity. The AUROC in Table 3 should be read as evidence that the score behaves sensibly on clinically annotated variation, not as an estimate of diagnostic accuracy.

Designed proteins are out of scope. Engineered fluorescent proteins score near zero at every model size. The method estimates evolutionary constraint, and sequences without evolutionary history have none to estimate.

Model capacity limits some families. Fast-evolving viral surface proteins scored near zero at 150M parameters, but influenza haemagglutinin recovers to rho 0.46 at 650M, indicating that this was a capacity limitation rather than a property of the approach.

Averaging discards direction. A position may tolerate conservative substitutions while forbidding drastic ones. The mean flattens this, which is why the interface also reports the extreme substitutions at each position.

Single chains only. AlphaFold monomer models are used, so residues that matter only within a complex are under-weighted.

Window placement. For proteins beyond 1,022 residues, scores depend marginally on where window boundaries fall.

Not a clinical tool. The method predicts evolutionary constraint, which correlates with but is not identical to pathogenicity. It is intended for hypothesis generation and must not inform diagnostic decisions.

6. Figures

A

DeepMutate-3D

Protein language model mutation scanning, painted onto the AlphaFold fold.

ESM-2 scores every possible point mutation as a log-likelihood ratio against the wildtype residue. Positions the model refuses to change are evolutionarily constrained, likely structural or functional cores, and glow **red**. Positions it happily swaps glow **blue**.

Example protein

Haemoglobin subunit alpha, P69905 (142 aa)

Protein sequence (FASTA or raw)

MVLSPADKTNVKAAWGKVGAHAGEYGAEALERMFLSFPTTKTYFPHFDLSHGSAGVKVGHGKGVADALTNVAHVDDMPNALSDLAHAKLRVDPVNFKLLSHCLLVTLAAHLPAEFTPAVHASLDKFLASVSTVLTISKYR

UniProt ID

Used for the AlphaFold structure, and to fetch the sequence.

P69905

Model

Larger models score better but cost more compute.

ESM-2 650M (most accurate)

Scoring mode

Masked marginals runs one forward pass per residue.

Masked marginals (accurate)

Wildtype marginals (fast)

Scan Mutations & Render 3D Fold

Sensitivity table (.csv) | Scored structure (.pdb)

Scan complete

- Residues scored: 142 | Device: cuda | Mode: Masked marginals (accurate)
- Model: facebook/esm2_t33_650M_UR50D | Inference time: 3.99 s (28.1 ms/residue)
- Mean sensitivity: -7.703 | Most fragile: D95 (-12.90), F44 (-12.47), H88 (-12.29), G60 (-11.79), G26 (-11.74)
- AlphaFold structure retrieved for P69905.

B

Residue Sensitivity Array

Pos	WT	Sensitivity	Verdict	Most damaging	Most tolerated
1	M	-6.283	Sensitive	M1W (-8.24)	M1A (-3.96)
2	V	-3.611	Tolerant	V2R (-7.66)	V2S (2.05)
3	L	-7.703	Sensitive	L3K (-9.83)	L3F (-4.58)
4	S	-7.637	Sensitive	S4W (-12.31)	S4T (-0.78)
5	P	-1.96	Tolerant	P5W (-7.88)	P5A (3.43)
6	A	-5.127	Sensitive	A6W (-9.96)	A6E (-0.59)
7	D	-10.465	Critical	D7W (-14.32)	D7E (-1.05)
8	K	-5.023	Sensitive	K8W (-9.15)	K8V (-1.21)
9	T	-3.982	Tolerant	T9W (-10.53)	T9A (2.13)
10	N	-1.165	Tolerant	N10W (-6.51)	N10A (2.64)
11	V	-8.676	Critical	V11H (-12.32)	V11I (1.13)
12	K	-4.754	Sensitive	K12W (-10.57)	K12Q (-0.94)
13	A	-5.882	Sensitive	A13W (-11.48)	A13S (-1.33)
14	A	-3.053	Tolerant	A14D (-7.08)	A14I (3.37)
15	W	-11.563	Critical	W15D (-15.44)	W15F (-4.46)

C

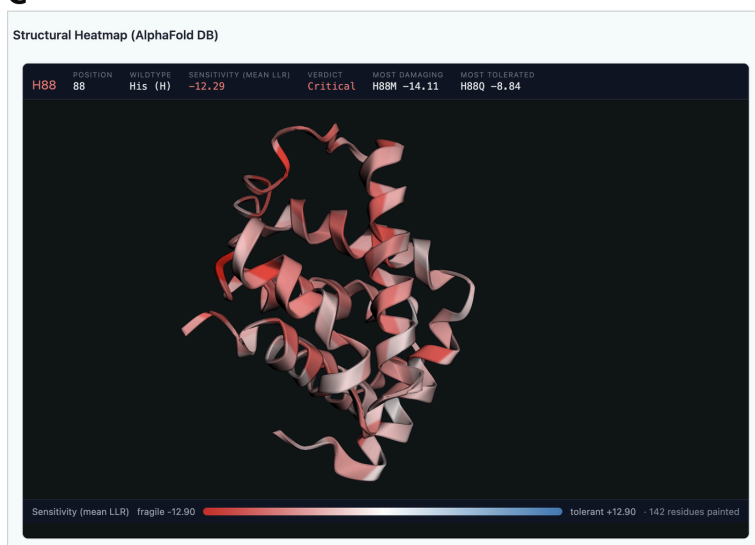


Figure 1. The DeepMutate-3D interface, scanning haemoglobin subunit alpha (UniProt P69905, 142 residues) with ESM-2 650M under masked-marginal scoring. (A) The complete interface: sequence and accession inputs, model and scoring controls, CSV and PDB downloads, and a summary reporting the device, model, inference time and the most constrained residues. The scan shown ran on dynamically allocated NVIDIA hardware in 3.99 s, 28.1 ms per residue. (B) The per-residue table, giving the sensitivity score, a categorical verdict, and the most damaging and most tolerated substitution at each position. (C) The AlphaFold model painted by residue sensitivity, with the hover readout for H88. UniProt annotates this position as the proximal haem b binding residue, and the scan ranks it third most constrained of 142 positions, in the top 2%, having seen only the sequence. Red marks constrained positions and blue substitution-tolerant ones, with the neutral midpoint fixed at zero so colour is comparable between proteins. Panels B and C are laid out as the application presents them.

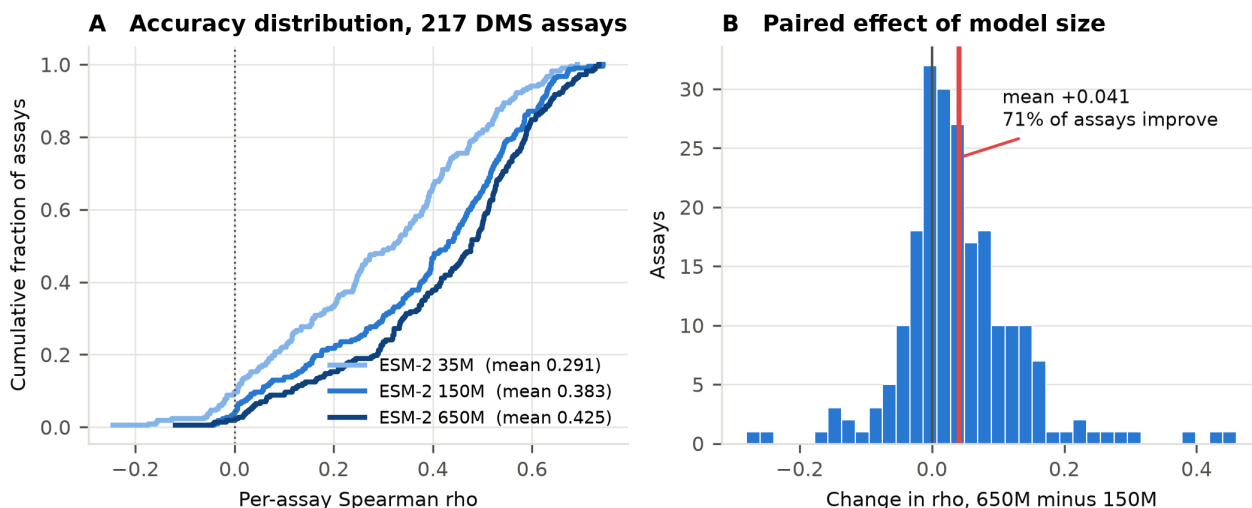


Figure 2. Effect of model size on ProteinGym. (A) Empirical cumulative distribution of per-assay Spearman correlation for three ESM-2 sizes across 217 assays. The distribution shifts uniformly rightward with model capacity rather than improving only on easy assays. (B) Paired per-assay difference between 650M and 150M. Pairing removes between-assay variance; the mean gain is +0.041 and 71% of assays improve.

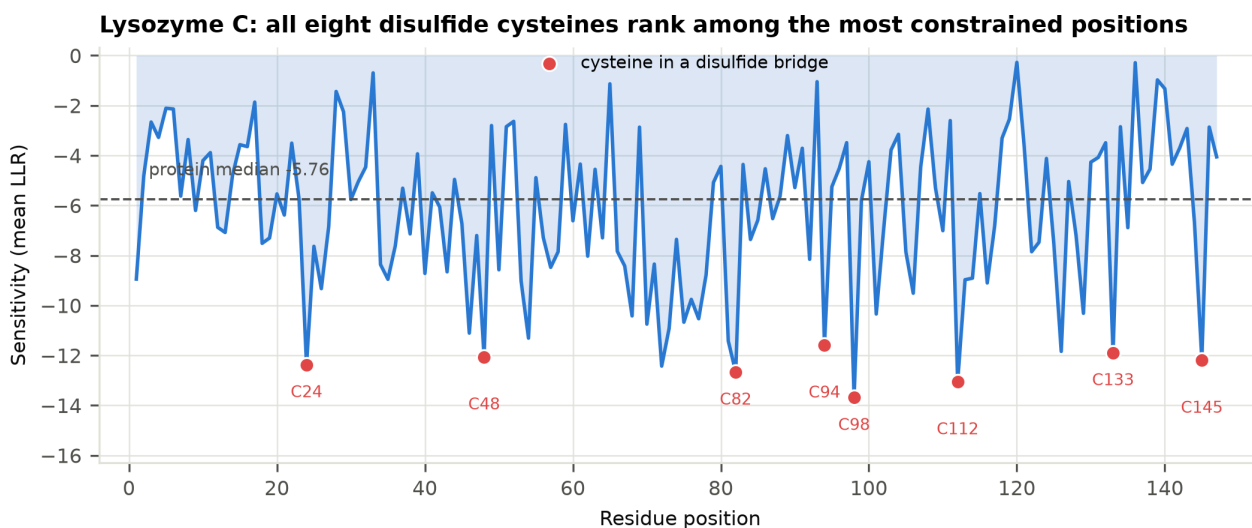


Figure 3. Sensitivity along the lysozyme C sequence. Per-residue sensitivity for all 147 positions, with the eight cysteines forming the four UniProt-annotated disulfide bridges marked. They occupy ranks 1, 2, 3, 5, 6, 7, 8 and 10 of 147. The dashed line is the protein median.

7. Availability

Source code, validation scripts, and all result files are at <https://github.com/Saadman/DeepMutate-3D> under Apache-2.0. The web application is at <https://huggingface.co/spaces/ras1992/DeepMutate-3D>. Every table in this note is reproducible from the scripts in `validation/` and `examples/`.

DeepMutate-3D retrieves but does not redistribute ESM-2 (MIT), AlphaFold structures (CC-BY-4.0), and UniProt sequences (CC-BY-4.0).

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